PROJECT REPORT

IoT-Based Smart Distribution Transformer Monitoring System

A low-voltage prototype for monitoring transformer parameters and providing automatic alerts

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Project type: Educational prototype

This report describes a low-voltage demonstration system, not a certified utility-grade transformer protection relay.

ABSTRACT

This project presents an Internet of Things (IoT)-enabled monitoring prototype for three separate single-phase transformer units. An ESP32 microcontroller collects electrical and temperature measurements from dedicated sensors. Readings can be displayed on an I2C LCD and remotely through a Blynk mobile dashboard. When a measured parameter crosses a configured threshold, the controller can command the corresponding relay output, activate that unit's cooling fan, and sound a common buzzer. An ultrasonic distance sensor may be used to estimate the level of a suitable tank when properly installed and calibrated. The prototype demonstrates remote monitoring, threshold-based alerts, and basic control logic. Sensor calibration, safe interfacing, and electrical isolation are essential before testing.

Keywords: ESP32, IoT, transformer monitoring, ACS712, ZMPT101B, DS18B20, Blynk, relay, LCD.

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1. INTRODUCTION

Distribution transformers reduce voltage to levels suitable for consumers and are important components of electrical power systems. Their condition can be affected by excessive loading, abnormal voltage, and high temperature. Periodic manual inspection may not provide continuous visibility of these parameters.

An IoT-based monitoring system uses sensors, a microcontroller, and a communication platform to collect and present operating information. In this project, the ESP32 reads sensor signals, updates a local LCD, sends measurements to Blynk, and applies threshold-based actions to relay, fan, and buzzer outputs.

1.1 Problem statement

The project aims to make voltage, current, and temperature readings easier to observe on a small educational transformer setup and to demonstrate a basic response when configured limits are exceeded.

1.2 Proposed solution

• Measure voltage and current for each of the three demonstration units.

• Measure temperature near each transformer using DS18B20 sensors.

• Display measurements locally and remotely using an LCD and Blynk dashboard.

• Demonstrate relay cut-off, unit-specific fan activation, and audible fault indication.

2. OBJECTIVES AND SCOPE

2.1 Objectives

• Develop an ESP32-based monitoring system for three separate single-phase transformer units.

• Monitor voltage, current, and temperature using dedicated sensor modules.

• Provide local display and remote viewing of readings and status.

• Demonstrate threshold-based protection alerts using relay outputs, fans, and a common buzzer.

• Use one OneWire data pin for all three uniquely addressed DS18B20 sensors.

2.2 Scope and assumptions

The demonstration setup uses three separate 230 V to 12 V, 3 A single-phase transformers rather than a direct 415 V three-phase transformer. The prototype uses individual cut-off outputs and does not perform automatic phase shifting. It is educational and is not a substitute for certified protection equipment.

3. SYSTEM OVERVIEW

The system has sensing, processing, communication, and output sections. Sensors provide signals to ESP32 inputs. The controller reads and calibrates the values, compares them with configured limits, updates the LCD, and publishes data to Blynk. When a fault condition is detected, it can command the affected unit's relay and fan and activate the common buzzer.

Functional overview — not a mains wiring diagram

Figure 1. Functional block diagram; this is not a detailed electrical wiring diagram.

3.1 Operating sequence

• Initialize the ESP32, sensors, LCD, outputs, Wi-Fi and Blynk connection.

• Read voltage, current, temperature and optional distance values.

• Check sensor readings for invalid or disconnected values.

• Compare current and temperature against configured thresholds.

• Apply the defined relay, fan and buzzer response if a fault is confirmed.

• Update the LCD and send readings/status to the mobile dashboard.

• Repeat the monitoring cycle at the programmed interval.

4. HARDWARE COMPONENTS

Component	Quantity / specification	Purpose
ESP32 development board	1; USB Type-C board	Controller, Wi-Fi and sensor processing.
Single-phase transformers	3 × 230 V/12 V, 3 A	Separate low-voltage demonstration units.
ACS712 current sensors	3 × 20 A modules	Current measurement; requires calibration.
ZMPT101B voltage sensors	3 modules	Voltage sensing with suitable calibration and safe installation.
DS18B20 temperature sensors	3 sensors; shared OneWire bus	Temperature measurement near each unit.
HC-SR04 ultrasonic sensor	1, optional	Distance measurement; level estimation only with suitable mounting.
I2C LCD	20×4; address often 0x27	Local display.
Relay module	4-channel module	Three unit outputs and one spare output.
DC cooling fans	3 × 5 V fans	Unit-specific cooling demonstration.
Common buzzer	1	Audible warning for configured faults.
Blynk IoT	Mobile dashboard	Remote readings and notifications when configured.

5. PROPOSED CONNECTIONS AND PIN ALLOCATION

This table lists the pin allocation used in the prototype. Confirm every connection against the actual hardware and the final Arduino sketch before energizing the circuit.

Function	GPIO / interface	Connection notes
ACS712 current — R unit	36	Analog input; calibrate offset and sensitivity.
ACS712 current — Y unit	39	Analog input; calibrate offset and sensitivity.
ACS712 current — B unit	34	Analog input; calibrate offset and sensitivity.
ZMPT101B voltage — R unit	35	Confirm output stays within ESP32 input limits.
ZMPT101B voltage — Y unit	32	Calibrate against a trusted meter.
ZMPT101B voltage — B unit	33	Calibrate against a trusted meter.
DS18B20 data	13	All three sensors share one OneWire bus; use correct pull-up and addresses.
HC-SR04 TRIG / ECHO	4 / 18	Optional; ECHO usually needs a divider/level shifter.
Relay — R / Y / B	23 / 26 / 27	Check relay module active logic and ratings.
Fan drivers — R / Y / B	16 / 17 / 19	Use suitable transistor/MOSFET driver circuits.
Common buzzer	25	Use a driver if buzzer current exceeds GPIO capability.
I2C LCD	SDA 21 / SCL 22	Confirm the exact ESP32 board mapping and voltage compatibility.
ESP32 power	USB Type-C	Use a regulated, board-compatible supply.

5.1 Safety precautions

• Never connect 230 V mains wiring to the ESP32 or a breadboard. Mains-side wiring must be enclosed, fused, strain-relieved and handled by a qualified person.

• A ZMPT101B module does not make every part of the circuit safe. Preserve isolation boundaries and use a suitable enclosure.

• Check relay contact ratings, isolation, coil supply and active logic. Software alone is not a safety disconnect.

• HC-SR04 ECHO commonly outputs 5 V; use a divider or level shifter before the ESP32 input.

• Use suitable supplies for fans and relay coils; protect DC loads against inductive spikes where required.

• Test with a low-voltage, current-limited setup first. Do not test with exposed mains wiring.

The GPIO list is a proposed firmware map, not a verified wiring diagram. The final circuit and source code must match.

6. WORKING PRINCIPLE AND ALGORITHM

6.1 Voltage measurement

Each voltage sensor module produces a signal related to the monitored AC voltage. The ESP32 samples the signal and converts it into a displayed value using an appropriate measurement method and calibration. A simple ADC multiplier may only provide an approximate value; calibrate against a trusted meter.

6.2 Current measurement

Each ACS712 module produces an output that varies with current. The controller converts the difference from the zero-current offset into current. Offset and sensitivity depend on the module and ADC behavior. A 20 A sensor rating does not mean the transformer or wiring can safely carry 20 A.

6.3 Temperature measurement

Three DS18B20 sensors can share one OneWire data pin because each sensor has a unique address. The firmware identifies the sensor for each transformer. Mount sensors securely and within their rated operating conditions.

6.4 Protection response

The prototype uses a current limit of 3.0 A and a temperature limit of 70 °C. These are prototype settings, not universal safe limits. When a limit is exceeded, the program can command the affected unit's relay to the defined safe state, turn on that unit's fan, activate the common buzzer, and publish the fault status. The final firmware must define whether manual reset is required.

6.5 Algorithm

1. Start ESP32 and initialize sensors, LCD, outputs and Blynk.

2. Read voltage, current and temperature values for each unit.

3. Validate sensor data and handle disconnected/invalid sensors as faults.

4. Compare readings with the configured thresholds.

5. If a fault is confirmed, command the appropriate relay/fan and common buzzer.

6. Update LCD and publish measurements/status to Blynk.

7. Repeat at the programmed monitoring interval.

6.6 Pseudocode

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For each transformer unit:
read voltage, current and temperature
update local and remote display
if current or temperature exceeds its limit:
command the unit relay to the defined safe state
turn on its fan and the common buzzer
publish fault status
otherwise follow the normal operating policy
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7. SOFTWARE AND IoT MONITORING

7.1 Arduino IDE

Arduino IDE is used to write, compile and upload the ESP32 firmware. Depending on the final sketch, libraries may include Blynk, LiquidCrystal_I2C, OneWire and DallasTemperature. Select the correct ESP32 board and COM port before uploading.

7.2 Blynk dashboard

The Blynk dashboard can display current, voltage and temperature for each unit, along with relay/fan/fault status. Phone notifications require correct event configuration in the Blynk console and firmware. Local LCD monitoring should continue even if the internet connection is unavailable.

Example virtual pin	Suggested data
V0–V2	Current readings for R, Y and B units
V3–V5	Voltage readings for R, Y and B units
V6–V8	Temperature readings for R, Y and B units
V9	Optional distance / level estimate
V10–V12	Unit status, relay state or fault indicator

The virtual-pin mapping above is illustrative. Use the exact mapping defined in the final Blynk template and sketch.

7.3 Security

• Never publish Blynk authentication tokens, Wi-Fi passwords or private credentials in the report, website, screenshots or public code repository.

• Hide credentials in screenshots and keep remote control access restricted.

• A public project webpage should contain explanation and safe diagrams, not secrets or unrestricted device-control credentials.

8. TESTING AND LIMITATIONS

8.1 Suggested test plan

Test	Procedure	Expected result
Power-up	Power ESP32 over USB and start firmware.	LCD initializes; Blynk connects if Wi-Fi is available.
Sensor baseline	Compare readings with a known safe reference condition.	Values are stable and close to reference after calibration.
Temperature response	Warm a sensor gently within its rating.	Displayed temperature changes; threshold logic responds.
Current threshold	Use a controlled low-voltage test load and calibrated meter.	Logic triggers at the configured threshold; do not intentionally overload unsafe equipment.
Relay/fan	Test outputs with mains safely isolated.	Correct unit output and fan respond as programmed.
Blynk display	Compare app readings with LCD.	Both show matching updated readings/status.
Sensor fault	Disconnect a sensor during a controlled test.	Firmware indicates invalid data rather than treating it as normal.

8.2 Limitations

• Sensor readings need calibration and may be affected by ADC noise, offsets, module tolerances and waveform distortion.

• Wi-Fi/cloud failure can interrupt remote updates; local logic should not depend on the internet.

• Ultrasonic distance is not a direct measurement of oil quality and only estimates surface distance.

• This prototype is not certified for unattended operation or direct use on a utility distribution transformer.

9. ADVANTAGES, APPLICATIONS AND FUTURE SCOPE

9.1 Advantages

• Combines important measurements in one monitoring system.

• Provides both local LCD display and remote mobile viewing.

• Demonstrates threshold-based audible and status alerts.

• Can command an appropriate relay and unit-specific cooling fan through suitable driver hardware.

• Provides a modular base for further study of electrical sensing and IoT.

9.2 Applications

• Electrical engineering laboratory demonstrations.

• Educational embedded-system and IoT projects.

• Prototype testing of sensor-based monitoring and alert logic.

9.3 Future scope

• Use calibrated true-RMS voltage/current measurement and document measurement uncertainty.

• Add local data logging, trend graphs and timestamped fault history.

• Add watchdog, sensor-failure detection and communication-loss indication.

• Develop a properly isolated sensing front-end and protected relay enclosure reviewed by a qualified professional.

• Use certified protection devices for real transformer applications.

10. CONCLUSION

The IoT-Based Smart Distribution Transformer Monitoring System demonstrates how an ESP32 can combine sensor measurements, local display, cloud monitoring and basic automatic response in a low-voltage educational setup. Voltage, current and temperature can be viewed locally and remotely, while configured threshold conditions can trigger a relay output, cooling fan and audible alert. The project supports practical learning in electrical measurement, embedded programming, IoT dashboards and control logic. Reliable operation depends on careful wiring, calibration, fault handling and safe separation between mains and low-voltage circuits. Professional review and further testing are required before real-world deployment.

END OF REPORT